A job ready bootcamp in C++, DSA and IOT

Strings



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Agenda

1. String Introduction
2. null character
3. User input
4. string functions
5. Function call by passing string
6. Handling multiple strings.

String

- String is a sequence of characters, terminated at null character.
- Strings are handled in char arrays.

```
char str[10];
```

Initializing char array during declaration

```
int main()
{
    char str[10] = { 'B', 'H', 'O', 'P', 'A', 'L' };
    int i;
    for(i=0; i<=9; i++)
     printf("%c", str[i]);

    return 0;
}
```

0	1	2	3	4	5	6	7	8	9
B	H	O	P	A	L	\0	\0	\0	\0

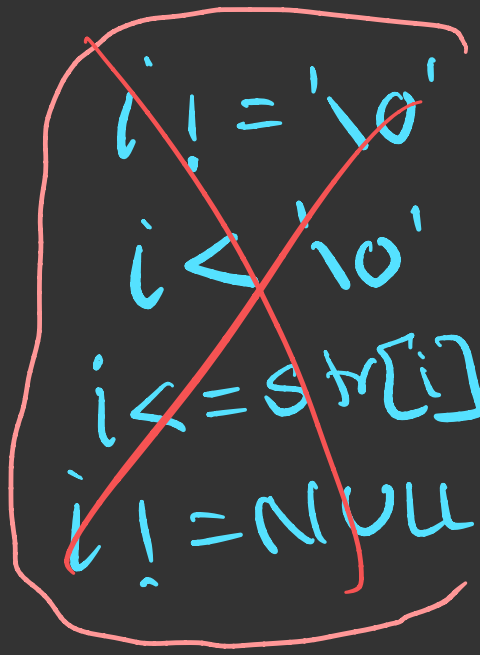
'\0'	0	
' '	32	
'A'	65	A
'B'	66	B

Printing String

```
int main()
{
    char str[10] = {'B','H','O','P','A','L'};
    int i;
    for(i=0;str[i] ;i++)
        printf("%c", str[i]);

    return 0;

}
```



i = 0, 1, 2, 3, 4 .....

str[i] = 'B', 'H', 'O', 'P' .....
..... '\0'

Improve your code

```
int main()
{
    char str[10] = { 'B', 'H', 'O', 'P', 'A', 'L' };
    int i;
    for (i=0; str[i]; i++)
        printf ("%c", str[i]);


}
```

Use of null character

```
int main()
{
    char str[10] = {'B','H','O','P','A','L'};

}
```

0	1	2	3	4	5	6	7	8	9
B	H	O	P	A	L	\0			

%s

```
int main()
{
    char str[10] = { 'B', 'H', 'O', 'P', 'A', 'L' };
    printf("%.s", str);
}
```

0	1	2	3	4	5
B	H	O	P	A	L
C	I	P	Q	B	M

i	str[i]
0	B
1	H
2	O

str[i] = str[i] + 1

'B' + 1

+ = 1

Calculating length of the String

```
int main()
{
    char str[10] = "BHOPAL";
    int i;
    for(i=0 ; str[i] ; i++);
    printf("Length is %d", i);

}
```

0 1 2 3 4 5 6

B H O P A L \0

String Constant

"BHOPAL" ← String constant
String Literal

```
char str[10] = "BHOPAL";
```

0	1	2	3	4	5	6	7	8	9
B	H	O	P	A	L	\0			

Taking input from User

- scanf is not capable to input multiword string scanf()
- because space, tab, new line characters are delimiters
- we will not use scanf for string input

gets()

- gets() is capable to input multiword string

gets()

only for strings

one string at a time

scanf()

multiple values

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↑ unreliable

fgets()

fgets(arrayname , inputsize , stdin)

Memory Concept

(byte = 8 bits

```
int i;
char str[20] = "My Str G";
```

option 1

```
for(i=0; str[i]; i++)
    printf("%c", str[i]);
```

option 2

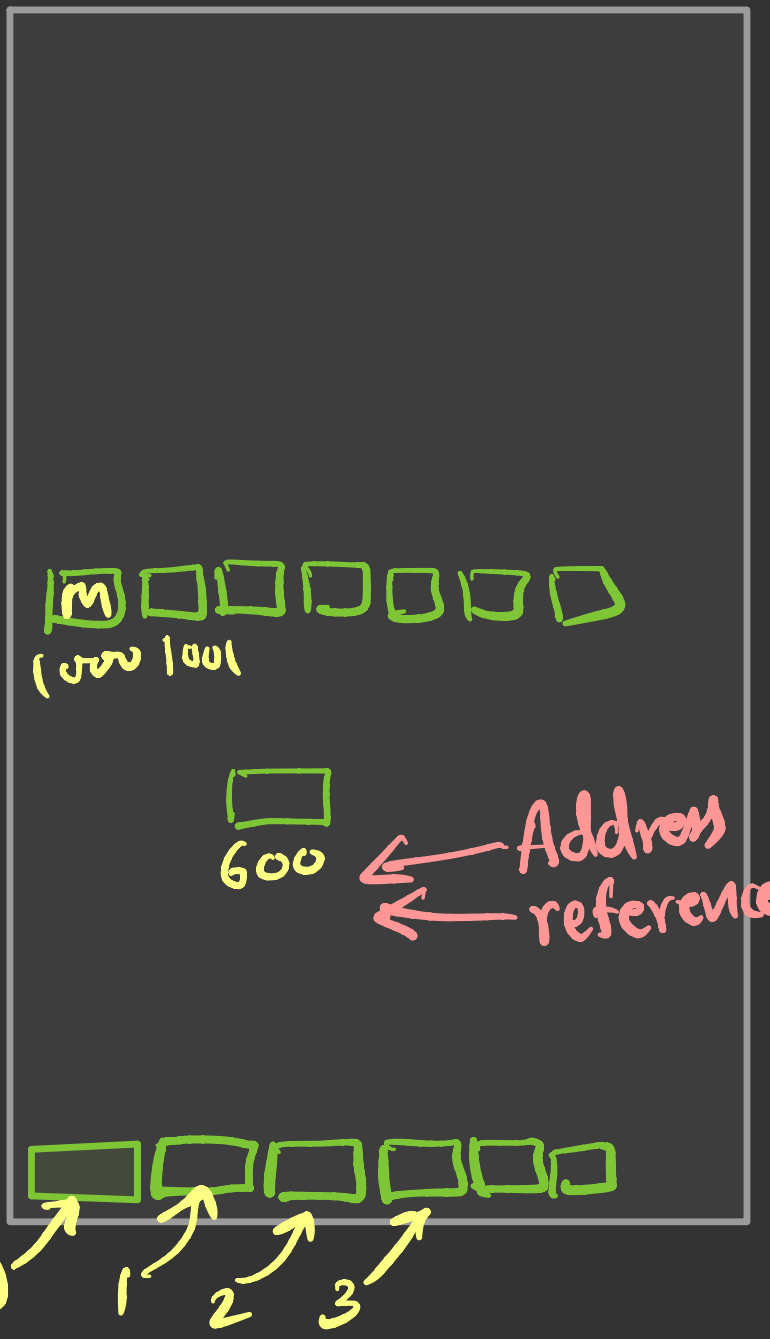
```
printf("%s", str);
```

&str[0] ≤ 1000
&str[1] ≤ 1001

```
char x;
```

str[0] 'M'

str == &str[0] 1000



String Functions

```
#include <string.h>
int    strlen(char *);
char*  strupr (char *);
char*  strlwr (char *);
char*  strrev (char *);
char*  strcpy (char*, char*);
char*  strcat (char*, char*);
int    strcmp (char*r, char*);
```

```
l = strlen(str);
l= strlen("Hello");
```

```
printf("%s", strupr(str));
strlwr(str);
strrev(str);
strcpy(str, "BHOPAL");
```

```
char s1[] = "Hello"
char s2[] = "Students"
strcat(s1, s2);
```

s1 → "Hello Students"

```
r = strcmp("Amit", "Amar");
```

r = -1 dictionary order

r = 0 equal

r = 1 opp of dictionary order

Function call By passing String

```
f1("Bhopal");
f1(str);
```

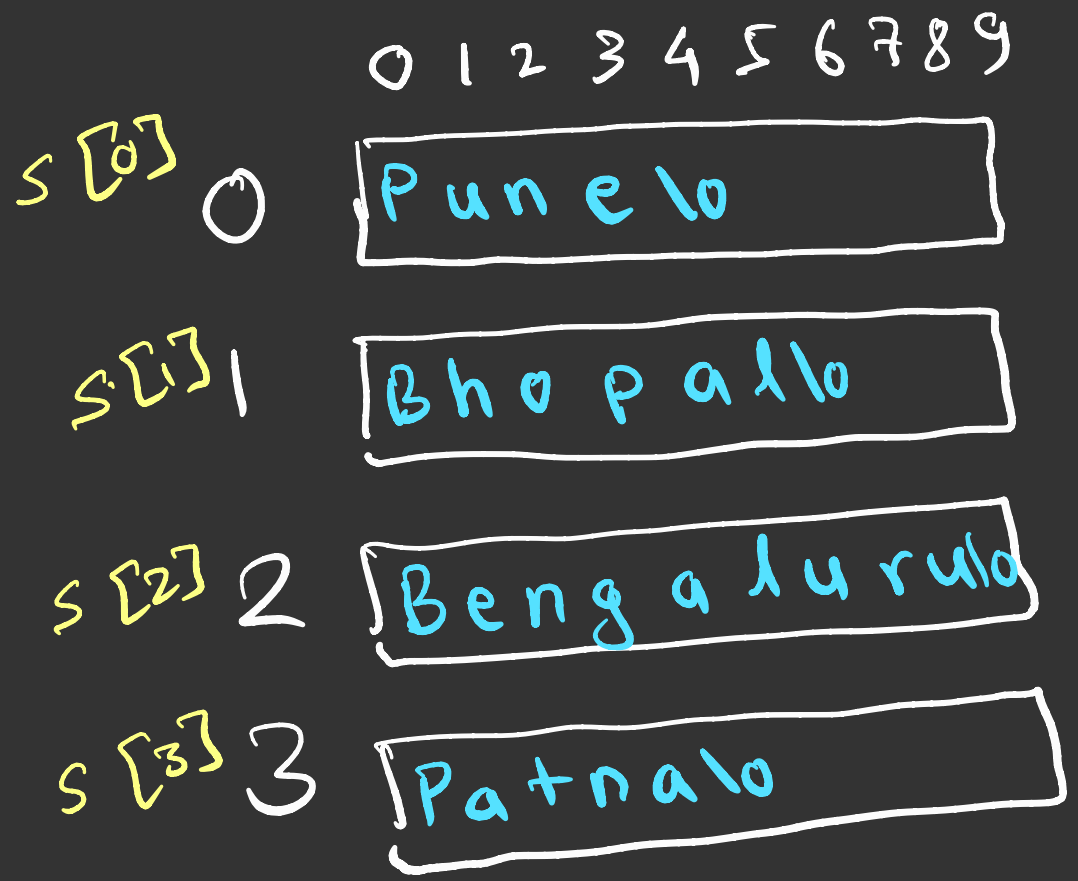
```
void f1(char s[])
{

}
```

Handling Multiple Strings

& s[0][0] == s[0]

```
char s[4][10] = { "Pune", "Bhopal", "Bengaluru",
                  "Patna" };
```




```
for(i=0; i<=3; i++)
    printf("%s", s[i]);
```

```
char s[5][20];
int i;
printf("Enter 5 city names");
for(i=0; i<=4; i++)
    fgets(s[i], 20, stdin);
```

Write a recursive function to print binary of a given decimal number.

```
void dtob(int x)
{
    if(x>0)
    {
        dtob(x/2);
        printf("%d", x%2);
    }
}
```

RC ① dtob(25) 11001

② dtob(25/2) 1100
printf("%d", 25%2); 1

BC ③ x>0

13

2451

2+4+5+1

① $n + sum(n-1)$

② $2*n-1 + sum(n-1)$

1	2	3	4	....	n
1	3	5	7	....	2n-1

③ $2*n + sum(n-1)$

④ $n*n + sum(n-1)$

⑤ $sum(n/10) + n\%10$

⑥ $n * fact(n-1)$

① Sum(2451) 12

② Sum(245) 11
+ 2451%10

⑦ HCF

(245, 125)

(245%125, 125)

(120, 125)

(120, 125%120)

(120, 5)

a b

if(a%b == 0)

return b;